



ELSEVIER

Available online at www.sciencedirect.com

SCIENCE @ DIRECT®

APPLIED
MATHEMATICS
AND
COMPUTATION

Applied Mathematics and Computation 170 (2005) 17–35

www.elsevier.com/locate/amc

Numerical studies on the split-step finite difference method for nonlinear Schrödinger equations

Hanquan Wang

Department of Computational Science, National University of Singapore, 10 Kent Ridge, Singapore 117543, Singapore

Abstract

In this study, we use the split-step finite difference method to solve various nonlinear Schrödinger equations including coupled ones. We study the numerical accuracy of the method. Detailed numerical results including higher dimensions show that the method provides accurate and stable solutions for nonlinear Schrödinger equations.

© 2004 Elsevier Inc. All rights reserved.

Keywords: Split-step finite difference method; Split-step Fourier spectral method; Nonlinear Schrödinger equations; Gross–Pitaevskii equation; Bose–Einstein Condensation

1. Introduction

Many problems in physics can be described by the nonlinear Schrödinger (NLS) equation [22]. One of the NLS equation with trapping potential, Gross–Pitaevskii equation attracts extreme interests recently. This equation

E-mail address: wanghanq@cz3.nus.edu.sg

describing the dynamics of Bose–Einstein Condensate at extremely low temperature (see [2,3] and reference inside), reads

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = -\frac{1}{2} \nabla^2 \psi + V_d(\mathbf{x}) \psi + \beta_d |\psi|^2 \psi, \quad \mathbf{x} \in \mathbb{R}^d, \quad t \geq 0, \quad (1.1)$$

where $V_d(\mathbf{x})$ is the trapping potential, it usually takes the form

$$V_d(\mathbf{x}) = \begin{cases} \frac{1}{2} x^2, & d = 1, \\ (x^2 + \gamma_y^2 y^2)/2, & d = 2, \\ (x^2 + \gamma_y^2 y^2 + \gamma_z^2 z^2)/2, & d = 3. \end{cases}$$

β_d is a real constant. The usual NLS equation does not include the trapping potential $V_d(\mathbf{x})$ [22].

Since analytical solutions for the NLS equation (1.1) is not available, one may need to seek its numerical solutions for possible physical applications such as Bose–Einstein condensation.

Moreover, when designing numerical schemes for (1.1) with the following initial conditions and boundary conditions,

$$\psi(\mathbf{x}, 0) = \psi^0(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^d, \quad (1.2)$$

$$\lim_{|\mathbf{x}| \rightarrow \pm\infty} \psi(\mathbf{x}, t) = 0, \quad t \geq 0 \quad (1.3)$$

two important invariants of (1.1), which need to be considered, are the *normalization of the wave function*

$$N(\psi) = \int_{\mathbb{R}^d} |\psi(\mathbf{x}, t)|^2 d\mathbf{x}, \quad (1.4)$$

and the *energy*

$$E(\psi) = \int_{\mathbb{R}^d} \left[\frac{1}{2} |\nabla \psi(\mathbf{x}, t)|^2 + V_d(\mathbf{x}) |\psi|^2 + \frac{\beta_d}{2} |\psi|^4 \right] d\mathbf{x}. \quad (1.5)$$

The problem now is how to design a stable and efficient numerical scheme for Eq. (1.1) as well as keeping their two conserved quantities. Recently Adhikari and coworkers (see [1] and reference inside) proposed a split-step Crank–Nicolson method for solving Eq. (1.1) with various geometries. This method first split the equation into a linear part and nonlinear part. For the linear equation we solve it using the Crank–Nicolson method in time and standard second order finite difference scheme in space. For the nonlinear part, we can solve it exactly. This method seems very efficient and stable, as it was pointed early by Weideman in [24] but he called this method as the split-step finite difference (SSFD) method. In this paper we will use this notation.

Except for the SSFD method, numerous numerical methods are proposed for solving NLS equation in the past years. For example, Weideman

et al. proposed a split-step fourier pseudospectral method [24]. Sulem proposed several numerical scheme including spectral method to study the singular solutions to the two-dimensional Cubic NLS equation [21]. Taha and Ablowitz compared eight numerical methods for one-dimensional NLS equation in [23]. Gardner et al. studied several numerical methods including finite element methods for one dimensional NLS equation in [14]. Chang et al., studied eight different schemes for the generalized one dimensional NLS equation in [8]. For more about the numerical methods for the NLS equation, see the review paper by Fairweather et al. [12].

For Eq. (1.1), Muruganandam et al. proposed one pseudospectral method based on Hermit functions [1]. Recently Perez–Garcia et al. studied several numerical methods for it in one dimension [19]. Bao et al. studied a time-splitting Fourier spectral approximation [2]. Cerimele et al. proposed an explicit finite difference method [7] and Lai et al. recently proposed an explicit Dufort–Frankel type scheme for the Gross–Pitaevskii equation of Bose–Einstein condensates on different geometries [18]. Dion et al. studied the Galerkin spectral based on Hermit functions [9].

Regarding the above-mentioned numerical methods, the split-step finite difference (SSFD) method proposed in [1,24] seems most favorite. In this paper we will apply this method to solve Eq. (1.1). This method have the advantages that it is easy to use and unconditionally stable. We will test its numerical accuracy for NLS equations of 1d, 2d and 3d. We will also test its numerical accuracy for a coupled NLS equations of 1d. and show that the method is unconditionally stable and keeps the conservation laws for these equation.

The paper is organized as follows. A detailed description of the SSFD method for the equation (1.1) will be given in Section 2. This method is then extended to solve coupled NLS equations. Detailed numerical algorithm is presented for the NLS equation of 1d and coupled NLS equations of 1d; in Section 3, To study the numerical accuracy of the SSFD method, we will study three test problems, a standard NLS equation of 1d which models many physical properties, a NLS equation with a period potential in 1d, 2d, and 3d, a coupled NLS equation of 1d appeared in Optics. In Section 4, Applications of SSFD method to Eq. (1.1) with a rotational term and coupled equations with a rotational term is presented. Detailed numerical results is given. In the last section, the conclusion will be made.

2. Split-step method

To introduce the method, let us first write Eq. (1.1) of 1d in the form

$$i \frac{\partial}{\partial t} \psi(x, t) = (A + B) \psi(x, t), \quad x \in R \quad (2.1)$$

where $A = -\frac{1}{2} \frac{\partial^2}{\partial x^2}$ is the linear operator and $B = V_1(x) + \beta|\psi(x, t)|^2$ is the nonlinear operator.

The main idea in the split step method, as mentioned before, for approximating the exact solution of Eq. (2.1) is solving the linear subproblem

$$i \frac{\partial}{\partial t} \psi(x, t) = A\psi(x, t), \quad (2.2)$$

and nonlinear subproblems

$$i \frac{\partial}{\partial t} \psi(x, t) = B\psi(x, t), \quad (2.3)$$

in a given sequential order.

For the first-order version of the split step method, the advancement in time is carried out in two steps. In the first step, an intermediate solution is computed by advancing the solution according to the purely nonlinear equation (2.3). In the second step, the solution is advanced according to the linear dispersive equation (2.2). To sum up, we may write the solution as the form

$$\psi(x, t + \Delta t) = e^{-iA\Delta t} e^{-iB\Delta t} \psi(x, t),$$

where Δt is the time step.

For the second-order version of the method, first solve (2.3), then use the solution as the initial condition, solve (2.2), at last use the solution as the initial condition, solve (2.3) and obtain the approximate solution. To sum up, we may obtain the solution as the form

$$\psi(x, t + \Delta t) = e^{-iB\Delta t/2} e^{-iA\Delta t} e^{-iB\Delta t/2} \psi(x, t), \quad (2.4)$$

this particular method is also called the standard Strang splitting. We may construct formulae of arbitrary order for the split step scheme [5,6,25]. After time-splitting, the nonlinear equation (2.3) can be solved exactly, while the linear equation (2.2) could be solved by using the Crank–Nicolson method in time and standard second order finite difference scheme in space. In the next two subsections, we will give the detailed algorithm for a single NLS equation and a coupled NLS equations in one dimension.

2.1. SSFD method for the NLS equation

For simplicity of notation, we shall first introduce the SSFD method in one space dimension ($d = 1$). We are now going to solve the following initial boundary value problem using the SSFD method:

$$i \frac{\partial \psi(x, t)}{\partial t} = -\frac{1}{2} \psi_{xx}(x, t) + V_1(x) \psi(x, t) + \beta |\psi|^2 \psi(x, t) \quad a \leq x \leq b, \quad t > 0, \quad (2.5)$$

$$\psi(x, t = 0) = \psi^0(x), \quad a \leq x \leq b, \quad (2.6)$$

$$\psi(a, t) = \psi(b, t) = 0, \quad t > 0. \quad (2.7)$$

According to the time-splitting method, we are now solving the linear subproblem

$$i \frac{\partial \psi(x, t)}{\partial t} = -\frac{1}{2} \psi_{xx}(x, t),$$

using Crank–Nicolson method in time and central difference in space. The non-linear subproblem $i \frac{\partial \psi(x, t)}{\partial t} = V_1(x) \psi(x, t) + \beta |\psi|^2 \psi(x, t)$ is solved exactly.

If we choose the spatial mesh size $h = \Delta x > 0$ with $h = (b - a)/M$ for M a positive integer, the time step $k = \Delta t > 0$ and let the grid points and the time step be

$$x_j := a + jh, \quad t_n := nk, \quad j = 0, 1, \dots, M, \quad n = 0, 1, 2, \dots$$

Let ψ_j^n be the approximation of $\psi(x_j, t_n)$ and ψ^n be the solution vector with components ψ_j^n .

We use the second version of split-step method and thus obtain the following algorithm of SSFD method for Eqs. (2.5)–(2.7):

$$\begin{aligned} \psi_j^* &= e^{-i(V_j + \beta |\psi_j^n|^2 k)/2} \psi_j^n, \quad j = 0, 1, 2, \dots, M, \\ i \frac{\psi_j^{**} - \psi_j^*}{k} &= -\frac{1}{4} \left[\frac{\psi_{j+1}^{**} - 2\psi_j^{**} + \psi_{j-1}^{**} + \psi_{j+1}^* - 2\psi_j^* + \psi_{j-1}^*}{h^2} \right], \\ j &= 1, 2, \dots, M-1, \\ \psi_0^{**} &= \psi_M^{**} = 0, \\ \psi_j^{n+1} &= e^{-i(V_j + \beta |\psi_j^{**}|^2 k)/2} \psi_j^{**}, \quad j = 0, 1, 2, \dots, M, \end{aligned} \quad (2.8)$$

the discretized equation (2.8) can be solved by Thomas' algorithm exactly [17], which is a direct solver and requires $O(M)$ operations.

In 2d, Eq. (1.1) becomes

$$\begin{aligned} i \frac{\partial \psi(x, y, t)}{\partial t} &= -\frac{1}{2} (\psi_{xx}(x, y, t) + \psi_{yy}(x, y, t)) + V_2(x, y) \psi(x, y, t) \\ &\quad + \beta_2 |\psi(x, y, t)|^2 \psi(x, y, t), \quad a < x < b, \quad c < y < d \end{aligned} \quad (2.9)$$

$$\psi(x, y, t = 0) = \psi^0(x, y), \quad a \leq x \leq b, \quad c \leq y \leq d \quad (2.10)$$

$$\psi(a, y, t) = \psi(b, y, t) = 0, \quad \psi(x, c, t) = \psi(x, d, t) = 0, \quad t > 0. \quad (2.11)$$

A SSFD method with the second order accuracy in both time and space for (2.9)–(2.11) is to solve the following equations with homogeneous boundary conditions subsequently:

$$i \frac{\partial \psi(x, y, t)}{\partial t} = V_2(x, y) \psi(x, y, t) + \beta_2 |\psi(x, y, t)|^2 \psi(x, y, t), \quad (2.12)$$

$$i \frac{\partial \psi(x, y, t)}{\partial t} = -\frac{1}{2} \psi_{xx}(x, y, t), \quad (2.13)$$

$$i \frac{\partial \psi(x, y, t)}{\partial t} = -\frac{1}{2} \psi_{yy}(x, y, t), \quad (2.14)$$

$$i \frac{\partial \psi(x, y, t)}{\partial t} = V_1(x, y) \psi(x, y, t) + \beta_2 |\psi(x, y, t)|^2 \psi(x, y, t), \quad (2.15)$$

where Eqs. (2.13) and (2.14) will be solved by Crank–Nicolson method in time and central difference in space, the corresponding discretized equations will be solved by Tom's algorithm. The detailed algorithm for solving the corresponding discretized equations is omitted here.

In 3d, a SSFD method with the second order accuracy in both time and space for (1.1) is to solve the following equations subsequently:

$$i \frac{\partial \psi(x, y, z, t)}{\partial t} = V_3(x, y, z) \psi(x, y, z, t) + \beta_3 |\psi(x, y, z, t)|^2 \psi(x, y, z, t), \quad (2.16)$$

$$i \frac{\partial \psi(x, y, z, t)}{\partial t} = -\frac{1}{2} \psi_{xx}(x, y, z, t), \quad (2.17)$$

$$i \frac{\partial \psi(x, y, z, t)}{\partial t} = -\frac{1}{2} \psi_{yy}(x, y, z, t), \quad (2.18)$$

$$i \frac{\partial \psi(x, y, z, t)}{\partial t} = -\frac{1}{2} \psi_{zz}(x, y, z, t), \quad (2.19)$$

$$i \frac{\partial \psi(x, y, z, t)}{\partial t} = V_3(x, y, z) \psi(x, y, z, t) + \beta_3 |\psi(x, y, z, t)|^2 \psi(x, y, z, t), \quad (2.20)$$

where Eqs. (2.17), (2.18) and (2.19) will be discretized by Crank–Nicolson method in time and central difference in space, the corresponding discretized equations will be solved by Tom's algorithm again.

Remark 2.1. After the discretization by the SSFD method, we can keep the norm (1.4) and the energy (1.5) unchanged numerically.

2.2. SSFD method for the coupled NLS equations

Since the SSFD method can be used to solve a single NLS equation, it is tempting to see whether the method can also be applicable to the coupled equation. In fact one can extend the above SSFD method for the coupled NLS

equations. For simplicity of notation, we shall introduce the SSFD method for the following coupled NLS equation:

$$i \frac{\partial \psi_j}{\partial t} = -\frac{1}{2} \frac{\partial^2 \psi_j}{\partial x^2} + V_j(x) \psi_j + i \alpha_j \frac{\partial \psi_j}{\partial x} + f_j(|\psi_1|^2, |\psi_2|^2) \psi_j, \quad j = 1, 2, \quad (2.21)$$

$$\psi_j(x, 0) = \psi_j^0(x), \quad j = 1, 2, \quad (2.22)$$

$$\psi_j(a, t) = \psi_j(b, t) = 0, \quad j = 1, 2. \quad (2.23)$$

There are two conserved quantities, i.e.,

$$\int_a^b \psi_j(x, t) dx = \int_a^b \psi_j^0(x) dx, \quad \text{for } j = 1, 2. \quad (2.24)$$

As for numerical approximation for this coupled NLS equations, earlier Bandrauk [5,6] have studied high-order split-step Fourier pseudospectral method for a coupled nonlinear Schrödinger equations of 1d. Later Ismail and Taha [15] designed an implicit finite difference method to solve this coupled nonlinear Schrödinger equations. Fewer numerical approximations for coupled NLS equations than a single NLS equation are found in the literature.

If we choose the spatial mesh size $h = \Delta x > 0$ with $h = (b - a)/M$ for M a positive integer, the time step $k = \Delta t > 0$ and let the grid points and the time step be

$$x_j := a + jh, \quad t_n := nk, \quad j = 0, 1, \dots, M, \quad n = 0, 1, 2, \dots$$

Let $\psi_{i,j}^n$ be the approximation of $\psi_i(x_j, t_n)$. The detailed algorithm of SSFD method for Eqs. (2.21)–(2.23) is

$$\begin{aligned} \psi_{1,j}^* &= e^{-i(V_{1,j} + f_1(|\psi_{1,j}^n|^2, |\psi_{2,j}^n|^2))k/2} \psi_{1,j}^n, \\ \psi_{2,j}^* &= e^{-i(V_{2,j} + f_2(|\psi_{1,j}^n|^2, |\psi_{2,j}^n|^2))k/2} \psi_{2,j}^n, \\ \psi_{1,j}^{**} &= -\frac{1}{4} \left[\frac{\psi_{1,j+1}^{**} - 2\psi_{1,j}^{**} + \psi_{1,j-1}^{**} + \psi_{1,j+1}^* - 2\psi_{1,j}^* + \psi_{1,j-1}^*}{h^2} \right] \\ &\quad - \frac{i\alpha_1}{4h} [\psi_{1,j+1}^{**} - \psi_{1,j-1}^{**} + \psi_{1,j+1}^* - \psi_{1,j-1}^*], \\ \psi_{2,j}^{**} &= -\frac{1}{4} \left[\frac{\psi_{2,j+1}^{**} - 2\psi_{2,j}^{**} + \psi_{2,j-1}^{**} + \psi_{2,j+1}^* - 2\psi_{2,j}^* + \psi_{2,j-1}^*}{h^2} \right] \\ &\quad - \frac{i\alpha_2}{4h} [\psi_{2,j+1}^{**} - \psi_{2,j-1}^{**} + \psi_{2,j+1}^* - \psi_{2,j-1}^*], \\ j &= 1, 2, \dots, M-1, \end{aligned}$$

$$\psi_{1,0}^{**} = \psi_{2,0}^{**} = \psi_{1,M}^{**} = \psi_{2,M}^{**} = 0,$$

$$\psi_{1,j}^{n+1} = e^{-i(V_{1,j} + f_1(|\psi_{1,j}^{**}|^2 + |\psi_{2,j}^{**}|^2))k/2} \psi_{1,j}^{**},$$

$$\psi_{2,j}^{n+1} = e^{-i(V_{2,j} + f_2(|\psi_{1,j}^{**}|^2 + |\psi_{2,j}^{**}|^2))k/2} \psi_{2,j}^{**}, \quad j = 0, 1, 2, \dots, M-1.$$

3. Numerical tests

In this section, we will test the numerical accuracy of the SSFD method through the following four examples. In Example 1, we will use the SSFD method to solve one standard NLS equation of 1d; in Example 2, we will use the SSFD method to solve another special NLSE with a period potential which recently appeared in Bose–Einstein condensation; in Example 3, we will use the SSFD to solve a coupled NLS equations of 1d which is used as a model in Optics. For numerical comparisons, we also show the numerical results for Examples 1–3 using the split-step Fourier spectral (SSFS) method proposed by Weideman [24,2]. Since the SSFS method can be also an efficient method for solving equations which have constant coefficients for the derivatives, this is true for these four examples.

Example 1. First we consider the following NLS equation in 1d,

$$i\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0$$

with homogeneous boundary conditions over $[-20, 20]$. The equation has the following analytical solution[14]

$$\psi_{\text{exact}}(x, t) = e^{i(2x-3t)} \text{sech}(x-4t),$$

We compute the maximum error between the exact solution and the approximated solution obtained, Table 1 gives the maximum of $|\psi_{\text{exact}}(x, t) - \psi_{\text{appro}}(x, t)|$ until time $t = 4.5$ using SSFD method and SSFS method. Table 2 shows that the method for this equation has second order accuracy in space.

Example 2. We consider the following (1.1) in 1d [10,11]

$$i \frac{\partial \psi}{\partial t}(x, t) = -\frac{1}{2} \psi_{xx} + V(x)\psi + |\psi|^2\psi, \quad x \in [0, 2\pi], \quad (3.1)$$

$$\psi_0(x) = \sin x, \quad (3.2)$$

where $V(x) = \cos^2 x$ is a periodic potential. The exact solution is $\psi_{\text{exact}}(x, t) = \sin x \exp(-i3t/2)$. Table 3 gives the maximum of $|\psi_{\text{exact}}(x, t) - \psi_{\text{appro}}(x, t)|$ until

Table 1

The maximum error between the exact solution and approximated solution at different time t in Example 2 for 1 d, $h = 0.01$, $\Delta t = 0.01$

t	SSFD	SSFS
1	3.862e-3	8.952e-5
2	8.173e-3	1.561e-4
3	1.291e-2	6.610e-4
4	3.6620e-2	3.663e-2
4.5	0.266e-1	0.266e-1

Table 2

The accuracy test for Example 1 by using SSFD method at $t = 2.0$, $\Delta t = 0.0001$

	$h = \frac{1}{4}$	$h = \frac{1}{8}$	$h = \frac{1}{16}$	$h = \frac{1}{32}$	$h = \frac{1}{64}$
Error	0.74661	0.17749	3.3686e-2	1.087e-2	2.7176e-3
Order	–	2.0726	2.0225	2.0057	2.001

time $t = 32.0$ using SSFD method and SSFS method. Table 4 also shows that SSFD method has second order accuracy in the space.

The test equation (1.1) in 2d is

$$i \frac{\partial \psi}{\partial t} = -\frac{1}{2}(\psi_{xx} + \psi_{yy}) + V(x, y)\psi + |\psi|^2\psi, \quad (x, y) \in [0, 2\pi] \times [0, 2\pi], \quad (3.3)$$

$$\psi_0(x, y) = \sin x \sin y, \quad (3.4)$$

where $V(x, y) = 1 - \sin^2 x \sin^2 y$. The exact solution for the problem is $\psi_{\text{exact}}(x, y, t) = \sin x \sin y \exp(-i2t)$. Table 5 gives the maximum of $|\psi_{\text{exact}}(x, y, t) - \psi_{\text{appro}}(x, y, t)|$ until time $t = 32.0$ using the SSFD method as

Table 3

The maximum error between the exact solution and approximated solution at different time t in Example 2 for 1 d, $h_x = h = 2\pi/128$, $\Delta t = 0.01$

t	SSFD	SSFS
4	4.057e-4	3.244e-12
8	8.115e-4	1.250e-11
12	1.217e-3	2.792e-11
16	1.623e-3	4.951e-11
20	2.029e-3	7.728e-11
24	2.434e-3	1.113e-10
28	2.840e-3	1.520e-10
32	3.246e-3	2.008e-10

Table 4

The accuracy test for the 1d problem in Example 2 by using SSFD method at $t = 2.0$, $\Delta t = 0.0001$, $h = h_x$

	$h = \frac{\pi}{4}$	$h = \frac{\pi}{8}$	$h = \frac{\pi}{16}$	$h = \frac{\pi}{32}$	$h = \frac{\pi}{64}$
Error	8.9249e-2	2.2661e-2	5.6872e-3	1.4232e-3	3.5588e-4
Order	–	1.9776	1.9944	1.9986	1.9997

Table 5

The maximum error between the exact solution and approximated solution at different time t in Example 2 for 2d, $h_x = h_y = 2\pi/128$, time step $\Delta t = 0.01$

t	SSFD	SSFS
4	8.115e-4	4.902e-12
8	1.623e-3	1.853e-11
12	2.434e-3	4.122e-11
16	3.246e-3	7.297e-11
20	4.057e-3	1.138e-10
24	4.867e-3	1.637e-10
28	5.680e-3	2.228e-10
32	6.492e-3	2.908e-10

well as the SSFS method. Table 6 shows that the method has second order accuracy in space.

The test problem in 3d is

$$i \frac{\partial u}{\partial t}(x, y, z, t) = -\frac{1}{2}(u_{xx} + u_{yy} + u_{zz}) + V(x, y, z)u + |u|^2 u, \quad (3.5)$$

$$(x, y, z) \in [0, 2\pi] \times [0, 2\pi] \times [0, 2\pi],$$

$$u_0(x) = \sin x \sin y \sin z, \quad (3.6)$$

where $V(x, y, z) = 1 - \sin^2 x \sin^2 y \sin^2 z$. The exact solution for the problem is $u(x, y, z, t) = \sin x \sin y \sin z \exp(-i5t/2)$.

Table 7 gives the maximum of $|\psi_{\text{exact}}(x, y, z, t) - \psi_{\text{appro}}(x, y, z, t)|$ until time $t = 32.0$ using the SSFD method as well as the SSFS method. Table 8 shows that the method has second order accuracy in space.

Table 6

The accuracy test for the 2D problem in Example 2 by using SSFD method at $t = 2.0$, $\Delta t = 0.0001$, $h = h_x = h_y$

	$h = \frac{\pi}{4}$	$h = \frac{\pi}{8}$	$h = \frac{\pi}{16}$	$h = \frac{\pi}{32}$	$h = \frac{\pi}{64}$
Error	0.3163	8.0329e-2	2.01604e-2	5.04497e-3	1.26155e-3
Order	–	1.9772	1.9944	1.9986	1.9997

Table 7

The maximum error between the exact solution and approximated solution at different time t in for the 3d problem in Example 2

t	SSFD	SSFS
4	1.217e-3	5.757e-12
8	2.434e-3	2.111e-11
12	3.651e-3	4.659e-11
16	4.869e-3	8.224e-11
20	6.086e-3	1.282e-10
24	7.303e-3	1.843e-10
28	8.520e-3	2.505e-10
32	9.737e-3	3.269e-10

$h_x = h_y = h_z = 2\pi/128$, time step $\Delta t = 0.01$.

Table 8

The accuracy test for the 3D problem in Example 2 by using SSFD method at $t = 2.0$, $\Delta t = 0.0001$, $h = h_x = h_y = h_z$

	$h = \frac{\pi}{4}$	$h = \frac{\pi}{8}$	$h = \frac{\pi}{16}$	$h = \frac{\pi}{32}$	$h = \frac{\pi}{64}$
Error	0.84044	0.21356	5.361e-2	1.3413e-2	3.3541e-3
Order	—	1.9765	1.9944	1.9986	1.9997

Example 3. We consider the test problem (2.21)–(2.23) where $f_1(x, y) = x + ey$ (e is a real constant), $f_2(x, y) = ex + y$, $V_1(x) = V_2(x) = 0$, and $-\alpha_1 = \alpha_2 = 0.5$. We can know that it admits one exact solution [15]

$$\psi_1^{\text{exact}}(x, t) = \sqrt{\frac{2\alpha}{1+e}} \text{sech}\left(\sqrt{2\alpha}(x - vt)\right) e^{i\left[(v-\delta)x - \left(\frac{v^2-\delta^2}{2} - \alpha\right)t\right]} \quad (3.7)$$

$$\psi_2^{\text{exact}}(x, t) = \sqrt{\frac{2\alpha}{1+e}} \text{sech}\left(\sqrt{2\alpha}(x - vt)\right) e^{i\left[(v+\delta)x - \left(\frac{v^2-\delta^2}{2} - \alpha\right)t\right]} \quad (3.8)$$

this solution describes the single soliton, the following parameter are used in our numerical calculation:

$$a = -20, \quad b = 80, \quad v = 0.1, \quad \delta = 0.5, \quad \alpha = 1.0, \quad e = \frac{2}{3}$$

we compute the error between the exact solution and the approximated solution obtained, Table 9 gives the maximum of $|\psi_1^{\text{exact}}(x, t) - \psi_1^{\text{appro}}(x, t)|$ at different time until $t = 32.0$ using SSFD method and SSFS method. In Table 10, we calculate $\epsilon_j = \|\psi_j^{\text{appro}}(x, t) - \psi_j^{\text{exact}}(x, t)\|_{L^2}$ ($j = 1, 2$) in order to test the accuracy of SSFD for the problem. From Table 6, we find the SSFD method (2.21)–(2.23) has second accuracy in space.

From Tables 3, 5, 7 and 9, we can see that the accuracy obtained from SSFD method is much lower than those obtained from SSFS method. This perhaps

Table 9

The maximum error between the exact solution and approximated solution at different time t for the equation in Example 3, $h = 0.05$, time step $\Delta t = 0.01$

t	SSFD	SSFS
4	9.843e-3	3.329e-4
8	1.849e-2	6.216e-4
12	2.796e-2	9.357e-4
16	3.752e-2	1.253e-3
20	4.643e-2	1.556e-3
24	5.601e-2	1.871e-3
28	6.536e-2	2.182e-3
32	7.442e-2	2.490e-3

Table 10

Numerical accuracy test result when $t = 2.0$, $\Delta t = 0.0001$ in Example 3

	$h_x = 1/2$	$h_x = 1/4$	$h_x = 1/8$	$h_x = 1/16$	$h_x = 1/32$	$h_x = 1/64$
ϵ_1	0.625	0.120	2.751e-2	6.750e-3	1.680e-3	4.195e-4
Order	–	2.3808	2.1250	2.0270	2.0064	2.0017
ϵ_2	0.606	0.111	2.531e-2	6.196e-3	1.541e-3	3.848e-4
Order	–	2.4488	2.1328	2.0303	2.0015	2.0017

due to the fact that the solved problem itself are periodic problems, and it is well known that Fourier pseudo-spectral method has spectral accuracy when applied into such kinds of problems with constant coefficients for the derivative terms. But we can observe that the SSFD method can be comparable to the SSFS method at large time using the same mesh size in both space and time.

As mentioned in the introduction, there are two conserved quantities for the single NLS equation: the norm and the energy of the complex function $\psi(\mathbf{x}, t)$. When using the SSFD method, we find that $|N(\psi^{\text{appro}}(\mathbf{x}, t)) - N(\psi^0(\mathbf{x}, t))| \leq 10^{-12}$, and $|E(\psi^{\text{appro}}(\mathbf{x}, t)) - E(\psi^0(\mathbf{x}, t))| \leq 10^{-4}$ until time $t = 32.0$ in Examples 1 and 2.

As mentioned before, there are two conserved quantities for the system of the coupled NLS Equations of 1d (2.21),(2.22),(2.23), the norms $\int_a^b |\psi_i(x, t)|^2 dx = \int_a^b |\psi_i(x, 0)|^2 dx$ ($i = 1, 2$). In Example 3, we find that $|\int_a^b |\psi_i^{\text{appro}}(x, t)|^2 dx - \int_a^b |\psi_i(x, 0)|^2 dx| \leq 10^{-12}$ from our numerical solution until time $t = 32.0$.

4. Numerical applications

In this section, in Example 4, we will apply the SSFD method to solve the equation (1.1) with a rotational term in two dimension.

In Example 5 and Example 6, we will also apply the SSFD method to solve two coupled NLS equations. One is Eqs. (2.21)–(2.23) with two different initial data; the other is the coupled NLS equation in two dimension.

Example 4. Eq. (1.1) of 2d with a rotational term

$$i \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = -\frac{1}{2} \nabla^2 \psi + V_2(\mathbf{x}) \psi + \beta_2 |\psi|^2 \psi - i\Omega(y\partial_x - x\partial_y) \psi, \\ \mathbf{x} \in \mathbb{R}^2, \quad t \geq 0, \quad (4.1)$$

which describe the dynamic properties of one-component Bose–Einstein condensates under external rotation ($\Omega > 0$). We now solve this equation using SSFD method, however the SSFS method is not applicable to solve this equation. In our numerical approximation, we will use the following initial data

$$\psi^0(x, y) = (\gamma_x \gamma_y)^{\frac{1}{4}} \exp^{-(\gamma_x x^2 + \gamma_y y^2)/2.0} / \pi^{\frac{1}{2}},$$

where $\gamma_x = \gamma_y = 1.0$ and solve this equation on the bounded domain $(x, y) \in [-8, 8] \times [-8, 8]$. Fig. 1 shows the time evolution of surface plot of $|\psi(x, y, t)|^2$ until $t = 1.0$ for $\Omega = 0.7$.

Example 5. The soliton interactions in Optics can be modelled by Eqs. (2.21)–(2.23). In order to simulate the interaction of two solitons, we again solve (2.21)–(2.23) with the following initial data [15]:

$$\psi_1(x, t) = \sum_{j=1}^2 \sqrt{\frac{2\delta_j}{1+e}} \operatorname{sech}(\sqrt{2\delta_j}) e^{i(v_j - \delta)x_j}, \quad (4.2)$$

$$\psi_2(x, t) = \sum_{j=1}^2 \sqrt{\frac{2\delta_j}{1+e}} \operatorname{sech}(\sqrt{2\delta_j}) e^{i(v_j + \delta)x_j}, \quad (4.3)$$

where $x_1 = x$, $x_2 = x - 25$. In our calculation, we set $a = -20$, $b = 80$, $v_1 = 1.0$, $v_2 = 0.1$, $\delta_1 = 1$, $\delta_2 = 0.5$, $e = 1$ and $-\alpha_1 = \alpha_2 = 0.5$. Fig. 2 describes the time evolution of the two solitons interactions at different times, $t = 10, 20, 30, 40, 50, 60$.

In order to simulate the interactions of three solitons, we also solve (2.21)–(2.23) with the following initial condition [15]:

$$\psi_1(x, t) = \sum_{j=1}^3 \sqrt{\frac{2\delta_j}{1+e}} \operatorname{sech}(\sqrt{2\delta_j}) e^{i(v_j - \delta)x_j} \quad (4.4)$$

$$\psi_2(x, t) = \sum_{j=1}^3 \sqrt{\frac{2\delta_j}{1+e}} \operatorname{sech}(\sqrt{2\delta_j}) e^{i(v_j + \delta)x_j} \quad (4.5)$$

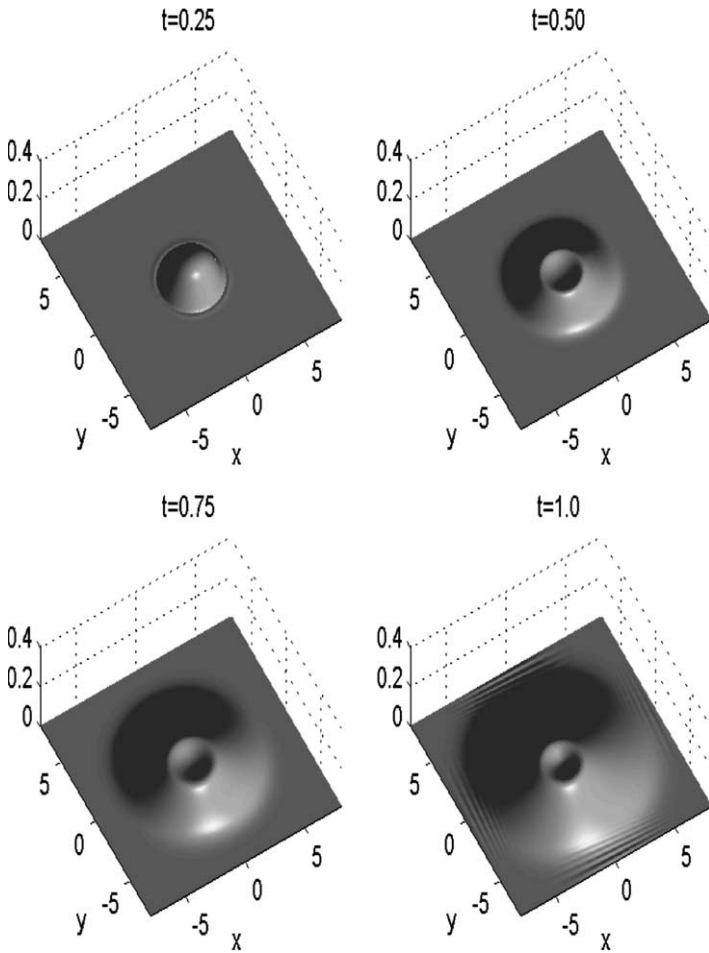


Fig. 1. Surface plot of $|\psi(x, y, t)|^2$ changes with time until $t = 1.0$ for Example 4.

where $x_1 = x$, $x_2 = x - 25$, $x_3 = x - 50$. In our calculations, we set $a = -20$, $b = 80$, $v_1 = 1.0$, $v_2 = 0.1$, $v_3 = -1.0$, $\delta_1 = 1.2$, $\delta_2 = 0.72$, $\delta_3 = 0.36$, $e = 1$. Fig. 3 describes the time evolution of the three soliton interactions at different times, $t = 10, 20, 30, 40, 50, 60$. Our figures (Figs. 2 and 3) are the same as those obtained in [15] when using same parameters in the numerical simulation.

Example 6. In Bose–Einstein Condensation, the coupled NLS equations

$$i \frac{\partial \psi_1}{\partial t} = \left(-\frac{1}{2} \nabla^2 + V_1(\mathbf{x}) + i\Omega \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) + \sum_{l=1}^2 \alpha_{1l} |\psi_l(\mathbf{x}, t)|^2 \right) \psi_1(\mathbf{x}, t), \quad (4.6)$$

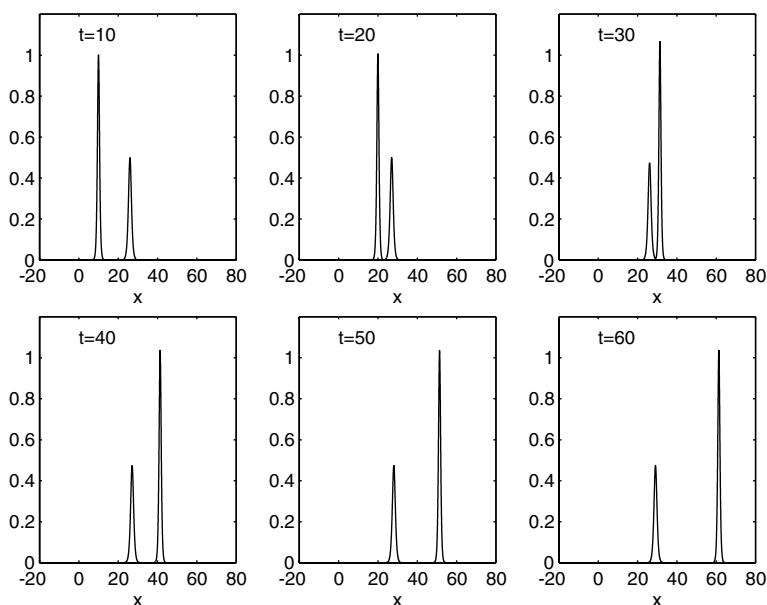


Fig. 2. The interactions between two solitons in Example 5, density $|\psi_1(x, t)|^2$ at different times.

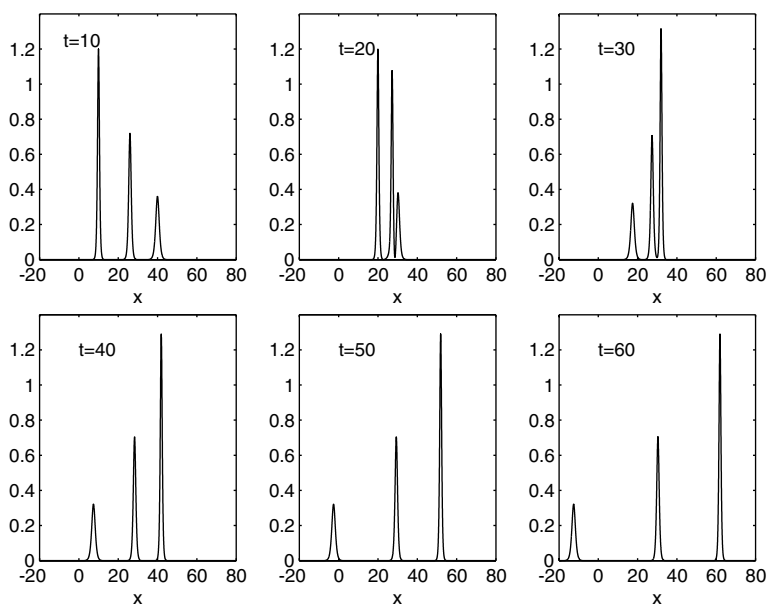


Fig. 3. The interactions between three solitons in Example 5, density $|\psi_1(x, t)|^2$ at different times.

$$i\frac{\partial\psi_2}{\partial t} = \left(-\frac{1}{2}\nabla^2 + V_2(\mathbf{x}) + i\Omega\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) + \sum_{l=1}^2 \alpha_{2l}|\psi_l(\mathbf{x},t)|^2 \right) \psi_2(\mathbf{x},t), \quad (4.7)$$

where $\alpha_{jk}, j, k = 1, 2$ are real constants. $V_i(\mathbf{x}) = \frac{1}{2}(w_{ix}^2 x^2 + w_{iy}^2 y^2)$, w_{ix}, w_{iy} ($i = 1, 2$) and Ω are real constants.

These equations describe the dynamic properties of quantized vortices in two-component Bose–Einstein condensates under external rotation [16].

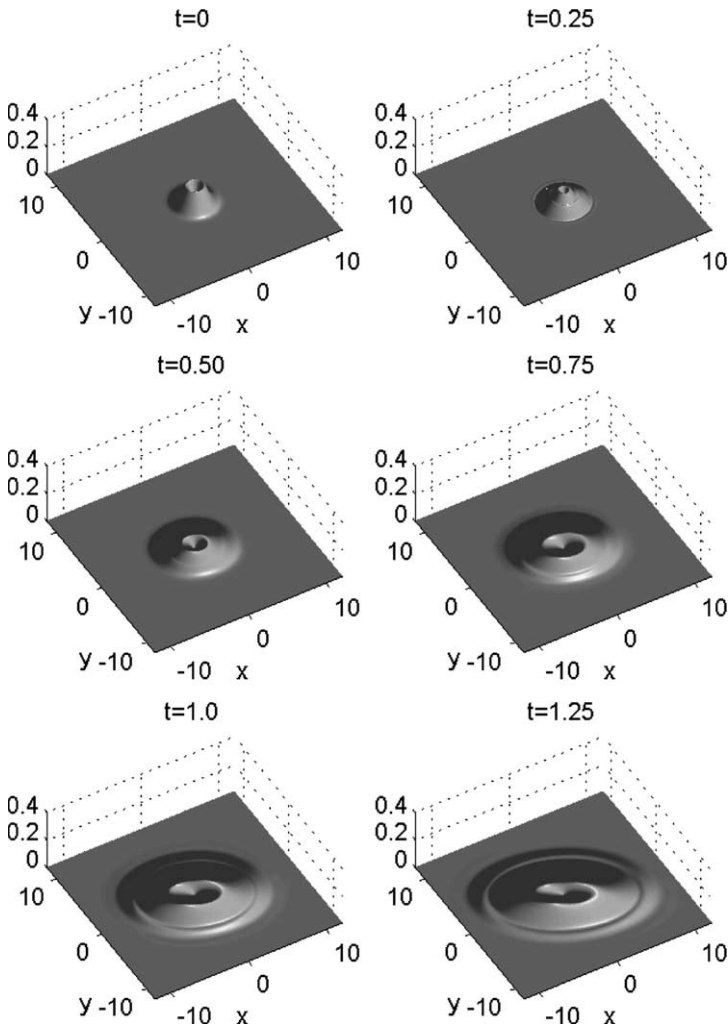


Fig. 4. Surface plot of $|\psi_1(x, y, t)|^2$ at different times until $t = 1.25$ for Example 6.

In the calculation we choose $(x, y) \in [-8, 8] \times [-8, 8]$, $\alpha_{jk} = 100$, $j = k = 1, 2$, $\alpha_{jk} = 80$, $j \neq k = 1, 2$, $\Omega = 0.7$ and $V_i(\mathbf{x}) = \frac{1}{2}(1.5x^2 + 0.5y^2)$ for $i = 1, 2$. We use the following initial conditions,

$$\psi_j(x, y, 0) = (x + iy) \exp^{-(\gamma_x x^2 + \gamma_y y^2)/2.0} / \pi^{\frac{1}{2}}, \quad j = 1, 2.$$

Figs. 4 and 5 shows surface plots of $|\psi_i(x, y, t)|^2$ ($i = 1, 2$) at different times until $t = 1.25$.

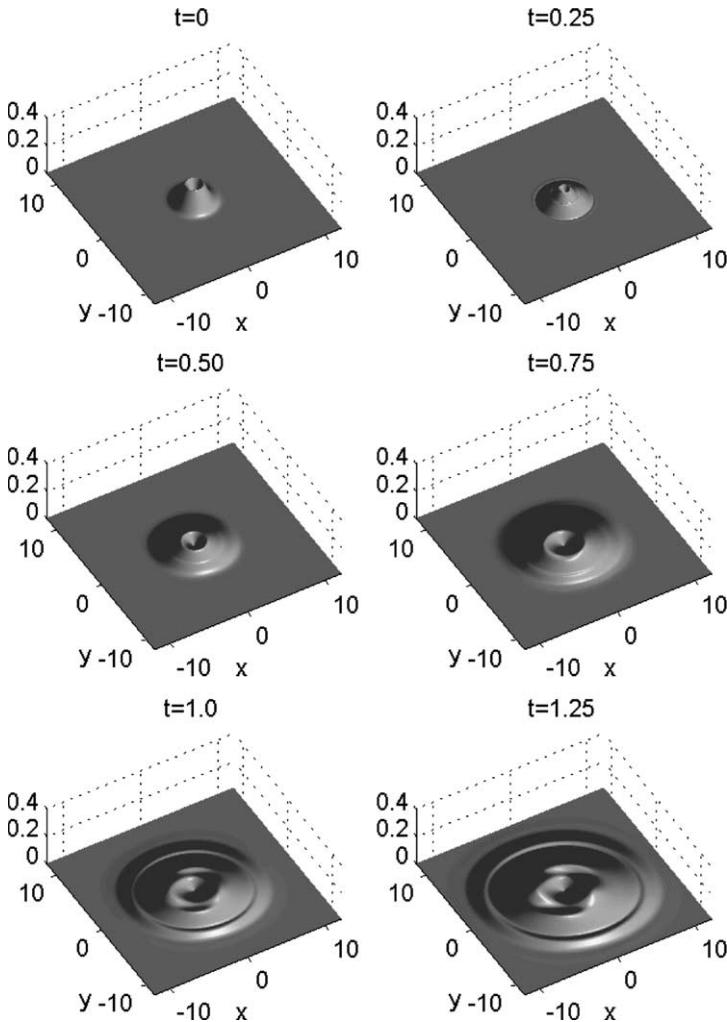


Fig. 5. Surface plot of $|\psi_2(x, y, t)|^2$ at different times until $t = 1.25$ for Example 6.

5. Conclusion

We test the numerical accuracy of the SSFD method for single NLS equation with trapping potential or the coupled NLS equations. Application of the SSFD method to the NLS equation with a rotational term is also presented. From our numerical test results shown in [Tables 1–10](#) the following observations for the SSFD method can be made: (a) The SSFD has second accuracy in space, even though without the spectral accuracy obtained by the SSFS method for some problems; (b) It is unconditionally stable; (c) It can be applied to solve more general problems such as problems with variable coefficients, and it is a direct solver and computational cost is relatively less than those using non-splitting method. (d) More important of all, it also keeps the conservation laws inherent in the single NLSE or the coupled NLS equations. Our numerical tests in 1d, 2d and 3d show that the method provide accurate solutions for the single NLS equation or the coupled NLS equations. In the future we plan to use this efficient SSFD method to study the dynamics of quantized vortices in Bose–Einstein Condensation. We will also justify analytically why the split-step method keep the conservation laws inside the NLS equation and coupled NLS equations.

Acknowledgement

The author thanks Prof. Weizhu Bao for valuable discussions and fruitful suggestions.

References

- [1] P. Muruganandam, S.K. Adhikari, Bose–Einstein condensation dynamics in three dimensions by the pseudospectral and finite-difference methods, *J. Phys. B* 36 (2003) 2501–2513.
- [2] W. Bao, D. Jaksch, P.A. Markowich, Numerical solution of the Gross–Pitaevskii equation for Bose–Einstein condensation, *J. Comput. Phys.* 187 (1) (2003) 318–342.
- [3] W. Bao, H. Wang, P.A. Markowich, Ground, symmetric and central vortex states in rotating Bose–Einstein condensates, Preprint.
- [5] A.D. Bandrauk, H. Shen, Exponential split operator methods for solving coupled time-dependent Schrödinger equations, *J. Chem. Phys.* 99 (2) (1993) 1185–1193.
- [6] A.D. Bandrauk, H. Shen, High-order split-step exponential methods for solving coupled nonlinear Schrödinger equations, *J. Phys. A-Math. Gen.* 27 (1994) 7147–7155.
- [7] M.M. Cerimele, M.L. Chiofalo, F. Pistella, S. Succi, M.P. Tosi, Numerical solution of the Gross–Pitaevskii equation using an explicit finite-difference scheme: an application to trapped Bose–Einstein condensates, *Phys. Rev. E* 62 (2000) 1382–1389.
- [8] Q. Chang, E. Jia, W. Sun, Difference schemes for solving the generalized nonlinear Schrödinger equation, *J. Comput. Phys.* 148 (1999) 397–415.

- [9] C.M. Dion, E. Cancès, Spectral method for the time-dependent Gross–Pitaevskii equation with a harmonic trap, *Phys. Rev. E* 67 (2003) 046706-1–046706-9.
- [10] B. Deconinck, B.A. Frigyi, J.N. Kutz, Stability of exact solutions of the defocusing nonlinear Schrödinger equation with periodic potential in two dimensions, *Phys. Lett. A* 283 (2001) 177–184.
- [11] B. Deconinck, B.A. Frigyi, J.N. Kutz, Dynamics and stability of Bose–Einstein condensates: the nonlinear Schrödinger equation with periodic potential, *J. Nonlinear Sci.* 12 (2002) 169–205.
- [12] G. Fairweather, M. Khebchareon, Numerical methods for Schrodinger-type problems, in: A.H. Siddiqi, M. Kočvara (Eds.), *Trends in Industrial and Applied Mathematics*, Kluwer Academic Publishers, Netherlands, 2002, pp. 219–250.
- [14] L.R.T. Gardner, G.A. Gardner, S.I. Zaki, Z. El Sahrawi, B-spline finite element studies of the non-linear Schrödinger equation, *Comput. Methods Appl. Mech. Eng.* 108 (1993) 303–318.
- [15] M.S. Ismail, R. Thiab, Taha numerical simulation of coupled nonlinear Schrödinger equation, *Math. Comput. Simul.* 56 (2001) 547–562.
- [16] K. Kasamatsu, M. Tsubota, M. Ueda, Structure of vortex lattices in rotating two-component Bose–Einstein condensates, *Physica B* 329–333 (2003) 23–24.
- [17] K.W. Morton, D.F. Mayers, *Numerical Solution of Partial Differential Equations*, Cambridge University Press, Cambridge, 1994.
- [18] M.C. Lai, C.Y. Huang, a simple Dufort–Frankel type scheme for the Gross–Pitaevskii equation of Bose–Einstein condensates on different geometries, preprint.
- [19] V.M. Perez-Garcia, X.Y. Liu, Numerical methods for the simulation of trapped nonlinear Schrödinger systems, *Appl. Math. Comput.* 144 (2–3) (2003) 215–235.
- [21] P.L. Sulem, C. Sulem, A. Patera, Numerical simulation of singular solutions to the two-dimensional cubic Schrödinger equation, *Commun. Pure Appl. Math.* 37 (1984) 755–778.
- [22] C. Sulem, P.L. Sulem, *The Nonlinear Schrödinger Equation, Self-focusing and Wave Collapse*, Springer, New York, 1999.
- [23] T.R. Taha, A numerical scheme for the nonlinear Schrödinger equation, *Comput. Math. Appl.* 22 (9) (1991) 77–84.
- [24] J.A.C. Weideman, B.M. Herbst, Split-step methods for the solution of the nonlinear Schrödinger equation, *SIAM J. Numer. Anal.* 23 (3) (1986) 485–507.
- [25] H. Yoshida, Construction of higher order symplectic integrators, *Phys. Lett. A* 150 (12) (1990) 262–268.